

Project Report

Summer Internship 2026

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Abstract

The objective of this internship was to study and explore Artificial Intelligence and Computer Vision technologies used in modern Virtual Try-On systems. The research focused on understanding body segmentation, pose estimation, garment fitting, avatar generation, and human body reconstruction techniques. Various AI models and frameworks were analyzed to understand their applications in both 2D and 3D Virtual Try-On systems. Practical exploration was performed using MediaPipe for pose detection and Blender for avatar visualization and asset exploration.

1. Introduction

The rapid growth of e-commerce has increased the demand for Virtual Try-On systems that allow users to visualize clothing before purchasing. Traditional online shopping platforms often lack personalization, making it difficult for users to understand how garments will fit or appear on their bodies.

Artificial Intelligence, Computer Vision, and 3D reconstruction technologies are enabling the development of advanced Virtual Try-On systems capable of generating realistic clothing visualizations. This internship focused on researching these technologies and understanding how they can be integrated to create next-generation virtual fashion experiences.

2. Objectives

The primary objectives of this internship were:

- Study Virtual Try-On technologies.
- Explore AI-based body segmentation methods.
- Analyze pose estimation techniques.
- Research garment mapping and fitting approaches.

- Investigate 3D avatar generation methods.
- Compare modern human body reconstruction frameworks.
- Explore future AI recommendation systems for fashion applications.
- Understand the role of AR and VR in virtual fashion.

3. Research Areas Covered

The internship covered the following research areas:

2D Virtual Try-On

- Garment overlay techniques
- Clothing alignment
- Body landmark detection
- Garment fitting approaches

Pose Detection

- MediaPipe
- OpenPose
- Detectron2

Body Segmentation

- Selfie Segmentation
- Detectron2 Segmentation

3D Avatar Generation

- SMPL
- PIFuHD
- ICON
- HumanNeRF
- 3D Gaussian Splatting

AI Recommendation Systems

- Personalized clothing recommendations
- Size prediction systems

- Fashion preference analysis

4. Datasets and Asset Collection

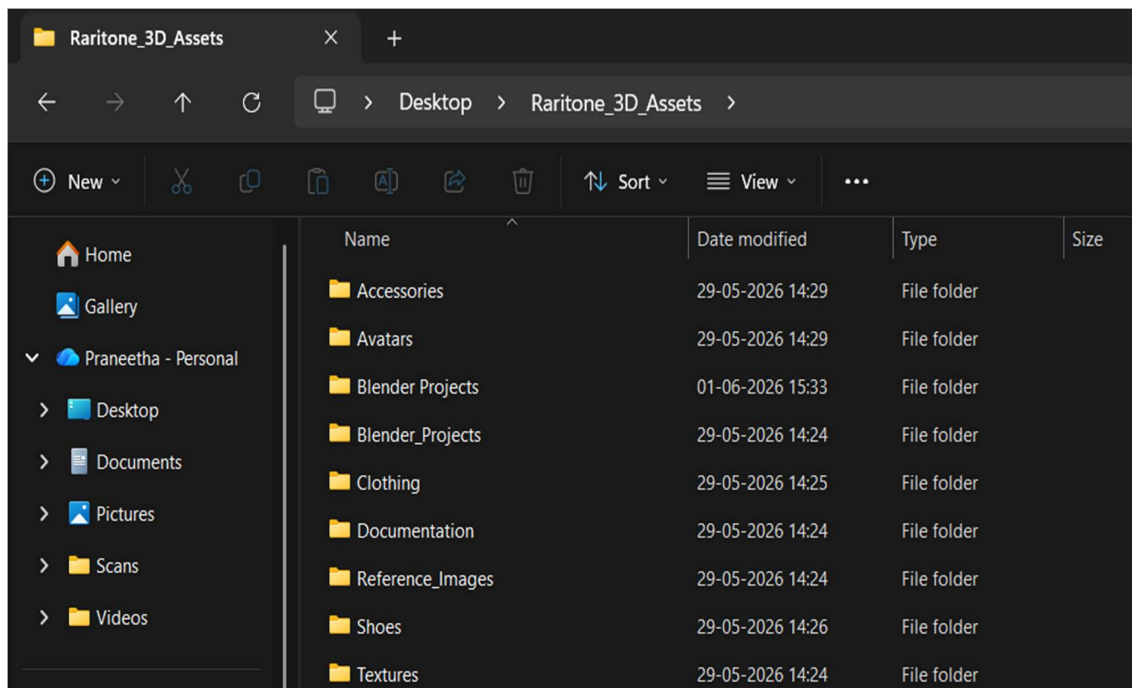
To support the research activities, various datasets and assets were collected and organized.

Dataset Categories

- Clothing Images
- T-Shirts
- Shoes
- Accessories
- Male Avatars
- Female Avatars
- 3D Models
- Reference Images

Asset Organization

The collected assets were organized into structured folders to improve accessibility and future usability. This repository can be utilized for future Virtual Try-On and avatar-based research projects.



5. 2D Virtual Try-On Research

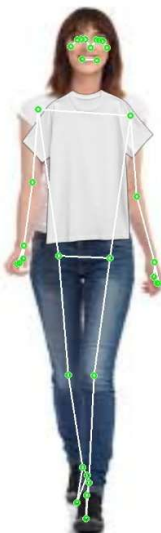
Research was conducted on traditional 2D Virtual Try-On systems that use image processing and pose estimation to overlay garments onto user images.

Activities Performed

- Garment overlay experiments
- Pose estimation using MediaPipe
- Clothing alignment testing
- Body landmark extraction
- Garment placement analysis

Findings

- MediaPipe provides fast and lightweight pose detection.
- Clothing alignment remains a major challenge.
- Different body poses significantly impact garment fitting quality.
- Improved segmentation can increase realism.



6. Body Segmentation and Pose Detection

Body segmentation and pose detection play a critical role in Virtual Try-On systems.

Models Studied

MediaPipe

- Fast landmark detection
- Real-time performance
- Lightweight implementation

OpenPose

- Accurate multi-person pose estimation
- Detailed body keypoint extraction

Detectron2

- Advanced segmentation capabilities
- High-quality object and body detection

Comparison

Model	Accuracy	Speed	Hardware Requirement
MediaPipe	High	Fast	Low
OpenPose	High	Medium	Medium
Detectron2	Very High	Slow	High

7. 3D Avatar Generation and Body Reconstruction

Modern Virtual Try-On systems increasingly rely on realistic digital avatars.

Technologies Studied

SMPL

A parametric human body model used for body shape and pose representation.

PIFuHD

A deep learning framework capable of generating detailed 3D human models from a single image.

ICON

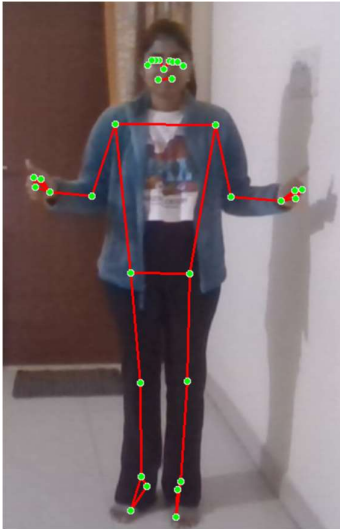
A reconstruction framework designed to improve clothed human reconstruction.

HumanNeRF

A neural rendering approach used for realistic human representation.

3D Gaussian Splatting

A modern 3D reconstruction technique that provides high-quality rendering and efficient visualization.



8. Blender Exploration

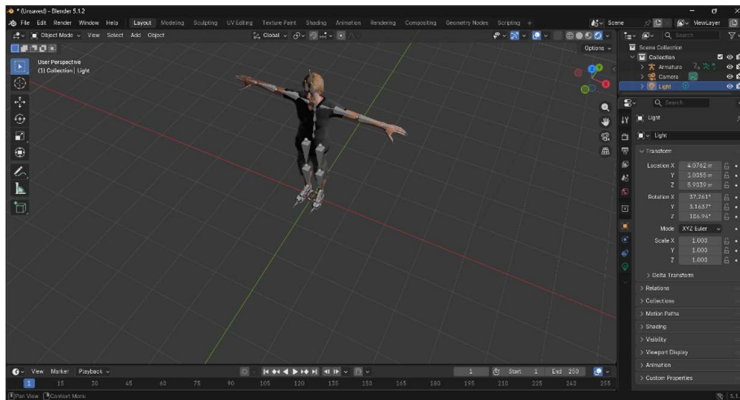
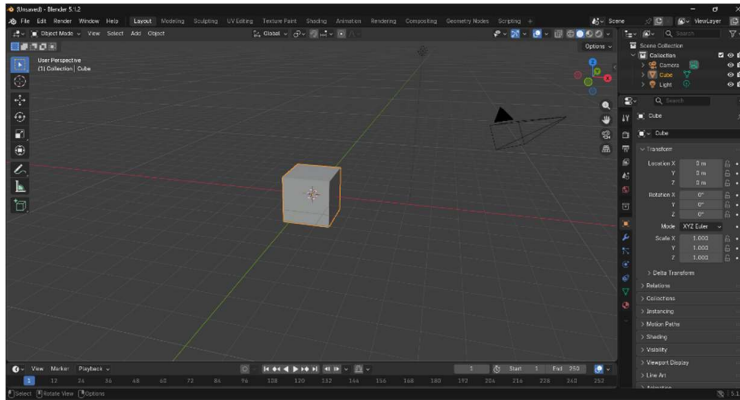
Blender was explored to understand avatar visualization and asset management workflows.

Activities Performed

- Imported avatar models
- Visualized 3D avatars
- Explored Material Preview mode
- Rotated and inspected avatar structures
- Conducted preliminary garment placement experiments

Observations

- Blender provides an effective platform for avatar visualization.
- 3D assets can be managed efficiently.
- Avatar-based systems offer significant advantages over traditional 2D approaches.



9. Complete AI Workflow

The overall Virtual Try-On workflow studied during the internship is shown below:

User Image



Body Detection



Body Segmentation



Pose Detection



Feature Extraction



Garment Mapping



Avatar Generation (Optional)



Virtual Try-On Result

10. Model Comparison

Technology	Purpose	Quality	Speed
MediaPipe	Pose Detection	High	Fast
OpenPose	Pose Detection	High	Medium
Detectron2	Segmentation	Very High	Slow
SMPL	Body Modeling	High	Fast
PIFuHD	Human Reconstruction	Very High	Slow
ICON	Clothed Human Reconstruction	Very High	Medium
HumanNeRF	Neural Rendering	Very High	Slow
Gaussian Splatting	3D Reconstruction	Very High	Fast

11. Challenges Faced

Several challenges were encountered during the research process:

- Clothing alignment issues
- Pose estimation inaccuracies
- Limited availability of high-quality datasets
- Complexity of advanced reconstruction frameworks
- Hardware limitations for large AI models

12. Solutions Applied

To address these challenges:

- Conducted comparative research on multiple frameworks.
- Explored Blender for practical understanding of avatars.
- Organized datasets systematically.
- Investigated future improvements in segmentation and reconstruction techniques.
- Studied emerging technologies for realistic avatar generation.

13. Future Improvements

Future work may include:

- CP-VTON implementation
- VITON-HD integration
- Detectron2-based segmentation
- OpenPose-based pose tracking
- SMPL avatar integration
- AR Virtual Try-On
- Real-time cloth simulation
- AI-based fashion recommendation systems

14. Key Learnings

Throughout the internship, significant knowledge was gained in:

- Artificial Intelligence
- Computer Vision
- Pose Detection
- Body Segmentation
- 3D Avatar Generation
- Human Reconstruction
- Blender Visualization
- Technical Research and Documentation

15. Conclusion

The internship provided valuable exposure to AI-powered Virtual Try-On technologies and modern computer vision systems. Through research, practical exploration, and comparative analysis, a strong understanding of body segmentation, pose estimation, avatar generation, and human reconstruction techniques was developed. The knowledge gained forms a strong foundation for future work in Virtual Try-On systems, virtual fashion, AR/VR applications, and AI-driven recommendation platforms.